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Soil Microbiome Needs Assessment Questionnaire

Interview Questions | Version 0.1

Contents

About this Interview....1
What to expect....2
Questions....2
- Scope and audience....2
- Sampling strategy....2
- Geographical & Site Context....3
- Physico-Chemical & Biological Context....3
- Specimen receipt and storage....4
- Specimen processing....4
- Library preparation....4
- Sequencing and Bioinformatics....5
- Data Management....5
- Associated Data Types....5
- Future Work (anticipating future needs)....6

Appendix A: Document Revision History....6

About this Interview

We are conducting a series of structured interviews with researchers working in soil microbiome genomics. The aim is to understand how contextual data are currently collected, managed, and shared, and what researchers need, in order to inform the design principles of a community data standard.

To get this right, we need to understand how researchers like you actually work: how samples are collected and tracked, what measurements are taken, how data is managed, and where the gaps and frustrations lie. This interview is one of a series being conducted across the research community, and your input will directly inform the structure and scope of the data model.

What to expect

The interview will take approximately 60–90 minutes and will be conducted as a structured conversation. The interviewer will guide you through a series of thematic areas, but the discussion may move fluidly between them depending on your project.

You do not need to prepare anything in advance, this document is purely for reference. The questions are open-ended and the interviewer may ask follow-up questions to get more specific detail about methods, tools, or units. This is to ensure answers are concrete enough to inform design decisions, not to probe beyond what you are comfortable sharing. In addition, you may not feel that the questions are relevant to your domain or project, that is fine and we can skip any question.

> *Privacy: Your responses will be treated as confidential. Findings will be anonymised and aggregated before informing any published outputs. If at any point you are uncertain whether something is sensitive, you are welcome to speak in general terms or ask us to note something off the record.*

Questions

Scope and audience

1. What are the primary questions or objectives of your project?
2. Who are your partners and/or collaborators?
3. What decisions (if any) will be informed by your data?
4. What land management practices (if any) are relevant to your project?
5. What level of comparability do you need? Is it within a project, across sites, across time, or across organisations?
6. Who ultimately consumes or interprets the data (e.g., modelers, regulators, farmers)?

Sampling strategy

7. Is there a sample plan? Is the sample plan, experimental design, or any other material accessible?
8. What kinds of samples are collected?
    a. Are only soil samples collected or are there additional sample types (e.g. host, environmental, food, wastewater)?
    b. What types of soil samples are specifically collected
9. What is your sample collection process?
    a. What is the sampling frequency and/or duration?
    b. What are your collection methods?

    c. What are your collection devices?

10. What unique identifiers (if any) are assigned to samples? (e.g. barcodes, lab IDs, field IDs, site-date-replicate codes)
    a. How are identifiers tracked across the full sample lifecycle?
    b. How are identifiers linked between parent samples and derived subsamples or aliquots?
11. Are you incorporating any previously collected samples or existing datasets?
12. Are you collecting any controls? What types?

Geographical & Site Context

13. What environmental site does your sampling area occupy?
14. What are the sites or land uses in the immediate vicinity that could influence your samples?
15. Is your site(s) exposed to any **planned** pre-sampling treatments or interventions (e.g. fertiliser, manure, pesticides/herbicides, mechanical events such as drilling, or experimental interventions)
16. Has your site(s) been exposed to any **unplanned** major environmental or anthropogenic events (e.g. fire, flood, drought, storm damage, animal intrusion)?
17. At what level of spatial specificity are sampling sites described? Is there a standardised approach to capture this information or is this recorded as a free text description/narrative?

Physico-Chemical & Biological Context

18. How is the soil classified or characterised in this project?
19. At what depth(s) are soil samples or measurements taken, and how are soil horizons or layers defined?
20. What physico-chemical data is being collected to describe the soil properties?
21. What nutrients or compounds are being measured?
22. Are there any other biological properties not already described that are measured beyond sequencing?
23. Are meteorological or microclimate data recorded?

Specimen receipt and storage

24. Are there formal receipt procedures when samples arrive at the lab?
25. Are environmental conditions during transport recorded?

26. How are samples stored and for how long?
27. How is contamination tracked and documented and at which stage is contamination most likely to occur? Is there a system to track sample integrity over time?

Specimen processing

28. Are samples further subsampled, pooled, or otherwise derived once received?
29. How is sample mass or volume quantified at each processing step? Is any normalisation applied?
30. Are the samples subjected to any special processing prior to nucleic acid extraction (e.g. fractionation, filtering, enrichment, culturing, etc)?
    a. How do processing steps potentially bias microbial composition?
    b. Is any culturing or isolation performed? If so, what media and growth conditions are used?
31. What DNA or RNA extraction methods are used?
32. How is the quality and quantity of extracted nucleic acid assessed (e.g. Qubit fluorometry, NanoDrop spectrophotometry, Bioanalyzer/TapeStation, gel electrophoresis)?
33. Are extraction protocols standardised across sites or labs?

Library preparation

34. What are the primary sequencing assay(s) being used (e.g WGS, WRS, amplicon, 16S, 18S etc)?
35. Is library preparation performed in-house or by an external partner or core facility?
36. What library preparation or enrichment methods are applied (e.g. tagmentation, ligation-based, PCR amplification, size selection)?
37. Are commercial library kits or in-house protocols used?
38. What primers are used?
39. Do you have a shareable or open access protocol?
40. What sequencing controls are included at the library preparation stage (e.g. mock community standards, synthetic spike-ins, endogenous controls, technical replicates)?

Sequencing and Bioinformatics

41. Is sequencing performed in-house or with a partner or external sequencing provider?
42. What sequencing instruments and platforms are used?

43. Can you walk through your bioinformatics workflow, from raw sequence data to final outputs?
44. What tools, reference databases, and custom scripts are used within the pipeline (e.g. Trimmomatic, DADA2, Kraken2, MetaPhlAn etc)?
45. How is taxonomy assigned, and which database is used?
46. Are metagenome-assembled genomes (MAGs) or binning approaches used?
47. How is quality control applied and recorded throughout the pipeline?
48. Is your workflow open access or shareable?

Data Management

49. How is the contextual data being captured (e.g. spreadsheets, LIMS, REDCap etc)?
50. Are any metadata or data standards applied to structure contextual data (e.g. MIxS/MIMARKS/MIMS checklists, Darwin Core, ISA-Tab, INSPIRE, DataCite)?
51. Where is the data being stored (e.g locally, public repositories, network databases)?
52. Is there a plan to submit data to a public repository?
53. What are the data sharing workflows - where does it need to go, how does it get there?
54. During data flow, will the data need to be transformed?
55. Are there governance, policy, or privacy considerations affecting data sharing?
56. Is there a formal Data Management Plan (DMP) in place?
57. How is data versioning or provenance tracked?

Associated Data Types

58. Are there additional data types associated with your samples that we have not yet discussed?
59. Which of these additional data types would be most useful to reference or flag alongside the primary sequencing data in a data standard?
60. Are there plans to formally link or integrate these datasets with the genomic data?

Future Work (anticipating future needs)

61. What data types would you want to prioritise collecting/integrating in the future?
62. What are your main pain points with contextual data collection or management today?
63. Are there emerging technologies or methods you anticipate adopting that might change your data needs?

64. Are there existing data standards or community resources that are close to meeting your needs but fall short in specific ways?

Appendix A: Document Revision History

Version	Date	Writer	Description of Change
0.1	March 2026	Charlie Barclay	Initial draft prepared